

SOFTWARE ENGINEERING & TESTING

ASSIGNMENT

Complete Questions and Comprehensive Answers Reference Manual

QUESTION 1

1. Create a decision table for a Flipkart-style coupon validation system where a coupon is only valid if: (1) the user is logged in, (2) the cart value is above ₹500, and (3) the coupon has not expired. Show all possible combinations and the resulting action (accept/reject).

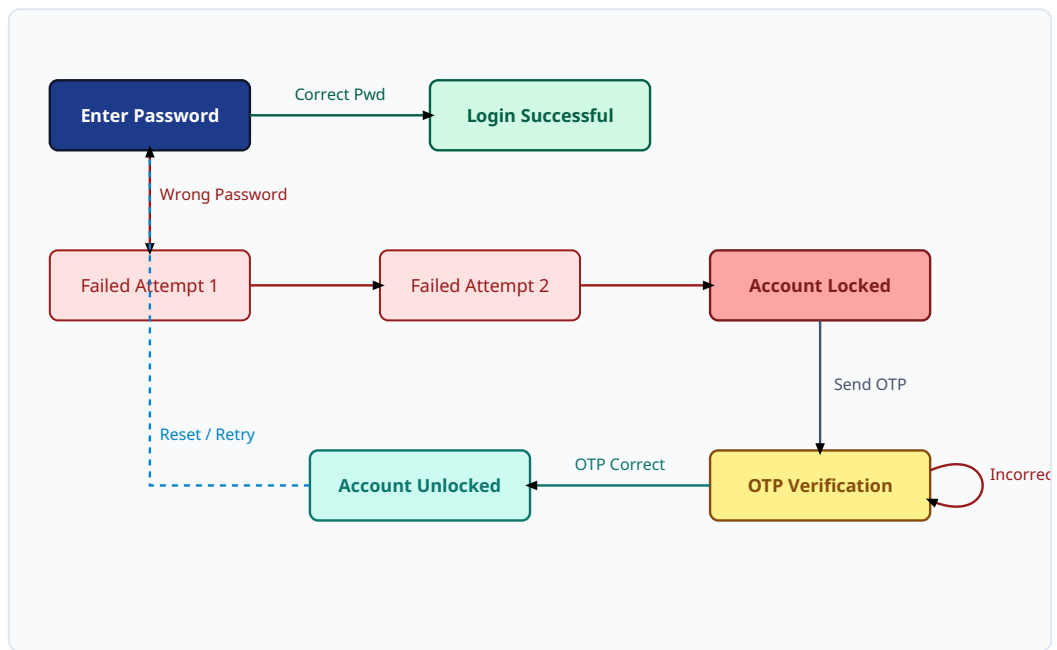
A complete matrix presenting all 8 possible truth combinations based on the 3 system validation parameters:

Conditions / Rules	R1	R2	R3	R4	R5	R6	R7	R8
User Logged In?	Y	Y	Y	Y	N	N	N	N
Cart Value > ₹500?	Y	Y	N	N	Y	Y	N	N
Coupon Not Expired?	Y	N	Y	N	Y	N	Y	N
Result Action	Accept	Reject	Reject	Reject	Reject	Reject	Reject	Reject

QUESTION 2

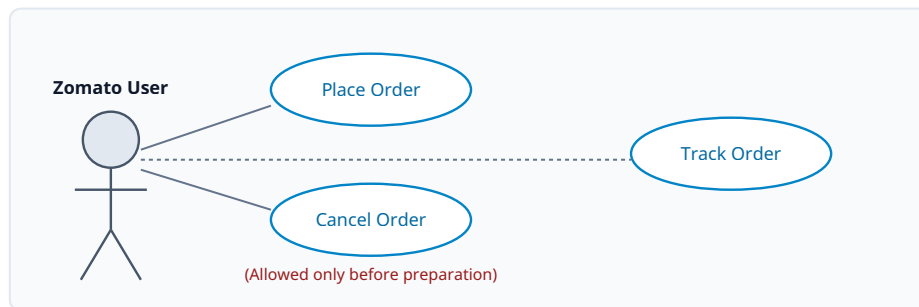
2. Draw a state transition diagram for a Paytm login feature that locks the account after 3 failed password attempts and only unlocks after a successful OTP verification.

The state flow diagrams map account status progression across sequential verification steps and failure loops:



QUESTION 3

3. Given the following use case: 'A Zomato user can place an order, cancel it before it is prepared, or track the order once prepared,' create a use case diagram and list at least 3 possible test scenarios based on it.



Test Scenario 1: Successful Order Placement

Precondition: User is authenticated; items added to checkout cart.

Steps: (1) Select cart validation page. (2) Execute absolute payment mechanism.

Expected Result: Order ID generates successfully, transaction logs record confirmation layout window.

Test Scenario 2: Cancel Order Before Preparation

Precondition: Active order state corresponds to "Pending acceptance" / "Not Prepared".

Steps: (1) Select active orders ledger. (2) Action the "Cancel Order" selection component.

Expected Result: Order processing ends; cancellation system dispatches instant notification alerts.

Test Scenario 3: Attempt to Cancel After Preparation

Precondition: Core business logic sets status to "Preparing".

Steps: (1) Load live order window tracking view. (2) Test termination button behavior.

Expected Result: Action blocked by system; returns alert: *"Order cannot be cancelled after preparation has started."*

QUESTION 4

4. Modify your decision table from Task 1 to include an additional rule: the coupon cannot be used more than twice by the same user. Update the table and explain how this new rule changes the possible outcomes.

Hint: Add a column for 'Coupon Usage Count' and adjust the actions accordingly.

Introducing the condition "**Coupon Usage Count ≤ 2**" doubles the logical permutations to a comprehensive 16-rule block:

Conditions / Rules	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16
User Logged In?	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N	N
Cart Value > ₹500?	Y	Y	Y	Y	N	N	N	N	Y	Y	Y	Y	N	N	N	N
Coupon Not Expired?	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N
Usage Count ≤ 2?	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
Result Action	ACC	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ	REJ

Behavioral Logic Impact Analysis:

By adding this condition checking sequence, the truth table doubles from 8 combinations to 16. Out of all potential tracks, **Rule 1 (R1) is the solitary configuration** that scores an **Acceptance** status. If a user triggers a match where the usage count restriction is violated (such as Rule 2), the transaction system will issue a rejection flag even if every single parameter from Task 1 is evaluated as perfectly true.